

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$$\mathbf{val}^\circ = (\mathbf{embed}^\circ \mathbf{wrap}) \cup (\mathbf{op}^\circ \neg \mathbf{Digit}(\mathbf{val}^\circ) \mathbf{snoc})$$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

the converse of $\mathbf{val}$, which reads digits as a number

(a)
(b)
(c)
(d)
(e)

(a)	$\mathbf{val}^\circ$	$\mathbf{val}$ undoes $\mathbf{wrap}$ or $\mathbf{snoc}$, reads the front with $\mathbf{val}$ and combines by $\mathbf{embed}$ or $\mathbf{op}$, so its converse runs that backwards
(b)	$= ([\mathbf{wrap}, \mathbf{snoc}]^\circ \mathbf{Digit}^{+-}(\neg \mathbf{xDigit}(\mathbf{val}))) [\mathbf{embed}, \mathbf{op}]^\circ$	undoing $\mathbf{wrap}$ or $\mathbf{snoc}$ first gives two cases: a decimal and its last digit read by $\mathbf{val}$ then $\mathbf{op}$, or one digit read by $\mathbf{embed}$
(c)	$= (\mathbf{wrap}^\circ \mathbf{embed})^\circ \cup (\mathbf{snoc}^\circ \neg \mathbf{xDigit}(\mathbf{val}) \mathbf{op})^\circ$	the converse of each case is its steps undone in reverse: undo $\mathbf{op}$, undo reading the front, then $\mathbf{snoc}$; or undo $\mathbf{embed}$, then $\mathbf{wrap}$
(d)	$= (\mathbf{embed}^\circ \mathbf{wrap}) \cup (\mathbf{op}^\circ \neg \mathbf{xDigit}(\mathbf{val})^\circ \mathbf{snoc})$	undoing the read of the front while keeping the last digit is $\mathbf{val}^\circ$ on the front alone
(e)	$= (\mathbf{embed}^\circ \mathbf{wrap}) \cup (\mathbf{op}^\circ \neg \mathbf{xDigit}(\mathbf{val}^\circ) \mathbf{snoc})$	

(6.1b)

- 1 $\mathbf{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$
 $\mathbf{op}^\circ$ splits a number into its quotient and remainder by 10
- 2 $(\exists p. 0 < \pi_1 p \wedge \mathbf{op}^\circ(m, p)) \iff 10 \leq m$
 $\mathbf{op}^\circ$ gives a pair with a nonzero first component exactly at the numbers with two or more digits
- 3 $(\exists d. \mathbf{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$
 $\mathbf{embed}^\circ$ gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

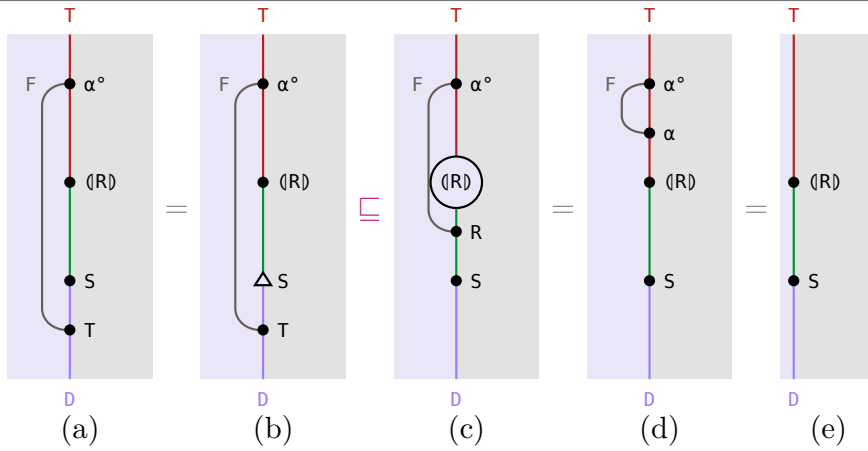
(6.2b)

the law	what it says
1 $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
2 $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
3 $\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
4 $X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

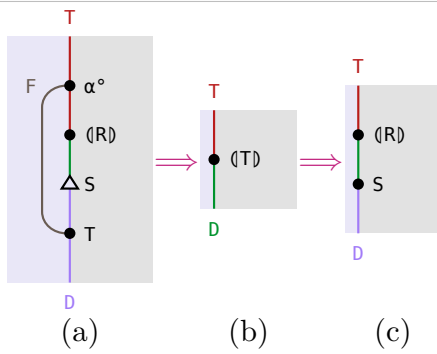
$$F(S) \ T \sqsubseteq R \ S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle \ S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S



(a)	$\alpha^\circ F(\langle R \rangle \ S) \ T$	
(b)	$= \alpha^\circ F(\langle R \rangle) \ F(S) \ T$	$F(S) \ T \sqsubseteq R \ S$
(c)	$\sqsubseteq \alpha^\circ F(\langle R \rangle) \ R \ S$	$\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$
(d)	$= \alpha^\circ \alpha \ \langle R \rangle \ S$	$\alpha^\circ \alpha = 1$
(e)	$= \langle R \rangle \ S$	



(a)	$F(S) \ T \sqsubseteq R \ S$	$F(S) \ T \sqsubseteq R \ S$ $\Rightarrow \alpha^\circ F(\langle R \rangle \ S) \ T \sqsubseteq \langle R \rangle \ S$
(b)	$\Rightarrow \alpha^\circ F(\langle R \rangle \ S) \ T \sqsubseteq \langle R \rangle \ S$	$\alpha^\circ F(X) \ R \sqsubseteq X$ $\Rightarrow \langle R \rangle \sqsubseteq X$
(c)	$\Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle \ S$	

$$R \ S \sqsubseteq F(S) \ T \Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$$

(6.2d)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



(a)	$\langle R \rangle \ S$	
(b)	$= \langle R \rangle \ S$	$\alpha^\circ \ \alpha = 1$
(c)	$= \alpha^\circ \ \alpha \ \langle R \rangle \ S$	$\alpha \ \langle R \rangle = F(\langle R \rangle) \ R$
(d)	$= \alpha^\circ \ F(\langle R \rangle) \ R \ S$	$R \ S \sqsubseteq F(S) \ T$
(e)	$\sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$	



(a)	$R \ S \sqsubseteq F(S) \ T$	$R \ S \sqsubseteq F(S) \ T$ $\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$
(b)	$\Rightarrow \langle R \rangle \ S \sqsubseteq \alpha^\circ \ F(\langle R \rangle \ S) \ T$	$X \sqsubseteq \alpha^\circ \ F(X) \ R$ $\Rightarrow X \sqsubseteq \langle R \rangle$
(c)	$\Rightarrow \langle R \rangle \ S \sqsubseteq \langle T \rangle$	

§6.3. Hylomorphisms

$$(\mathbb{S})^\circ (\mathbb{R}) = (\mu X : S^\circ F(X) R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



(a)	$S^\circ F((\mathbb{S})^\circ (\mathbb{R})) R$	
(b)	$= S^\circ F((\mathbb{S})^\circ) F((\mathbb{R})) R$	$\alpha (\mathbb{R}) = F((\mathbb{R})) R$
(c)	$= S^\circ F((\mathbb{S})^\circ) \alpha (\mathbb{R})$	$(\mathbb{R})^\circ \alpha^\circ = R^\circ F((\mathbb{R})^\circ)$
(d)	$= (\mathbb{S})^\circ \alpha^\circ \alpha (\mathbb{R})$	$\alpha^\circ \alpha = 1$
(e)	$= (\mathbb{S})^\circ (\mathbb{R})$	



(a)	$(\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R$	$(\mathbb{R})^\circ \alpha^\circ = R^\circ F((\mathbb{R})^\circ)$
(b)	$= S^\circ F((\mathbb{S})^\circ ((\mathbb{S})^\circ \setminus X)) R$	$S (S \setminus R) \subseteq R$
(c)	$\sqsubseteq S^\circ F(X) R$	$S^\circ F(X) R \subseteq X$
(d)	$\sqsubseteq X$	

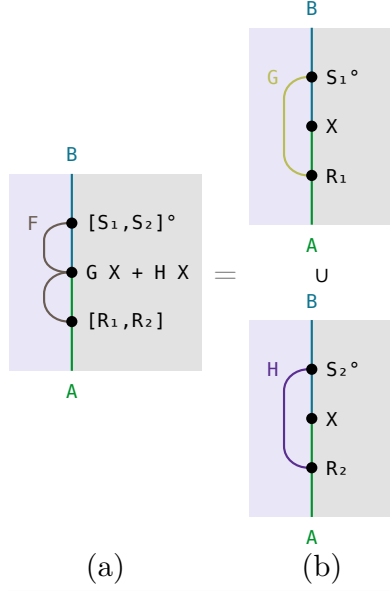


(a)	$S^\circ F(X) R \subseteq X$	$S^\circ F(X) R \subseteq X$
(b)	$\Rightarrow (\mathbb{S})^\circ \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \subseteq X$	$X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$
(c)	$\Leftrightarrow \alpha^\circ F((\mathbb{S})^\circ \setminus X) R \subseteq (\mathbb{S})^\circ \setminus X$	$\alpha^\circ F(X) R \subseteq X$
(d)	$\Rightarrow (\mathbb{R}) \subseteq (\mathbb{S})^\circ \setminus X$	$X \subseteq S \setminus R \Leftrightarrow S X \subseteq R$
(e)	$\Leftrightarrow (\mathbb{S})^\circ (\mathbb{R}) \subseteq X$	

$$(\forall x. \forall y. \forall X. F(X) = G(X) + H(X)) \Rightarrow ([S_1, S_2]^\circ \llbracket [R_1, R_2] \rrbracket = (\mu X : (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2))) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

the body of Theorem 6.2 at $F(X)=G(X)+H(X)$

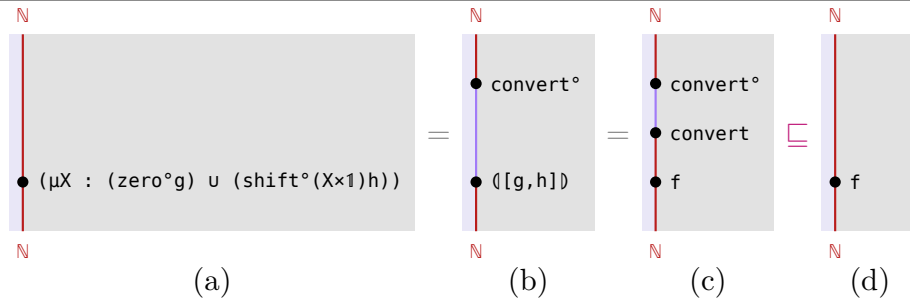


(a)	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	$[S, T] = (S+T)\nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the + stands for the codiagonals ∇°, ∇ , which are not drawn
(b)	$= (S_1^\circ G(X) R_1) \cup (S_2^\circ H(X) R_2)$	$\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h)) \sqsubseteq f \quad (6.4a)$$

once f sends **zero** to g and turns **shift** into h , the recursion that halves the argument at each step computes f



(a)	$(\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$	$\text{convert}^\circ \llbracket [g, h] \rrbracket = (\mu X : (\text{zero}^\circ g) \cup (\text{shift}^\circ (X \times 1) h))$
(b)	$= \text{convert}^\circ \llbracket [g, h] \rrbracket$	$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h$ $\Rightarrow \text{convert } f = \llbracket [g, h] \rrbracket$
(c)	$= \text{convert}^\circ \text{convert } f$	$\text{Simple}(\text{convert})$
(d)	$\sqsubseteq f$	

$$(\mu X : (\text{zero}^\circ \text{one}) \cup (\text{shift}^\circ (X \times 1) \text{op}(a))) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes a to the power b

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d

$$(\mu X : (\text{zero}^\circ \text{zero}) \cup (\text{shift}^\circ (X \times 1) \text{op}(b))) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves a at each step computes the remainder of a by b

$\text{zero } \text{mod}(b) = \text{zero}$

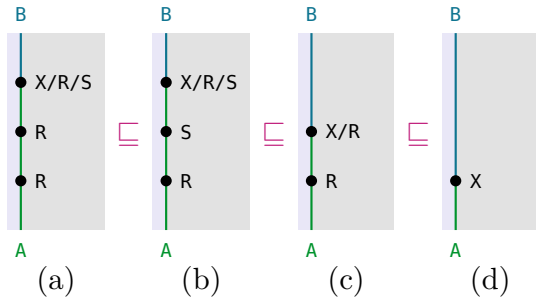
$0 \text{ mod } b$ is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

$(2a+d) \text{ mod } b$ is $\text{op}(b)$ of $a \text{ mod } b$ and d

§6.5. Unique fixed points

$$\begin{aligned}
 & \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{X} & (6.5a) \\
 & ((\mathbf{X}/\mathbf{R})/\mathbf{S}) \text{ followed by } \mathbf{R}\mathbf{R} \text{ stays below } \mathbf{X} \text{ when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \\
 & \mathbf{inductive}(\mathbf{S}) \wedge \mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R} \Rightarrow \mathbf{inductive}(\mathbf{R}) \\
 & \mathbf{R} \text{ is inductive when } \mathbf{R}\mathbf{R} \sqsubseteq \mathbf{S}\mathbf{R} \text{ for an inductive } \mathbf{S}
 \end{aligned}$$



(a)	$(\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{R} \mathbf{R}$	$\mathbf{R} \mathbf{R} \sqsubseteq \mathbf{S} \mathbf{R}$
(b)	$\sqsubseteq (\mathbf{X} / \mathbf{R} / \mathbf{S}) \mathbf{S} \mathbf{R}$	$(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$
(c)	$\sqsubseteq (\mathbf{X} / \mathbf{R}) \mathbf{R}$	$(\mathbf{R} / \mathbf{S}) \mathbf{S} \sqsubseteq \mathbf{R}$
(d)	$\sqsubseteq \mathbf{X}$	

$$\mathbf{member} = \emptyset \quad (6.5b)$$

a constant relator records no elements, so its membership is empty

$$\mathbf{member} = [\mathbf{member}, \mathbf{member}]$$

a member of $\mathbf{F}+\mathbf{G}$ is a member of whichever summand the value is in

$$\mathbf{member} = (\pi_1 \mathbf{member}) \cup (\pi_2 \mathbf{member})$$

a member of $\mathbf{F} \times \mathbf{G}$ is a member of either component

$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{X} = \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \wedge \mathbf{Y} = \mathbf{S} \mathbf{F}(\mathbf{Y}) \mathbf{R} \Rightarrow \mathbf{X} = \mathbf{Y} & (6.5c) \\
 & \text{when } \mathbf{S} \mathbf{member} \text{ is inductive, two solutions } \mathbf{X}, \mathbf{Y} \text{ of } \mathbf{X} = \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R} \text{ are equal}
 \end{aligned}$$

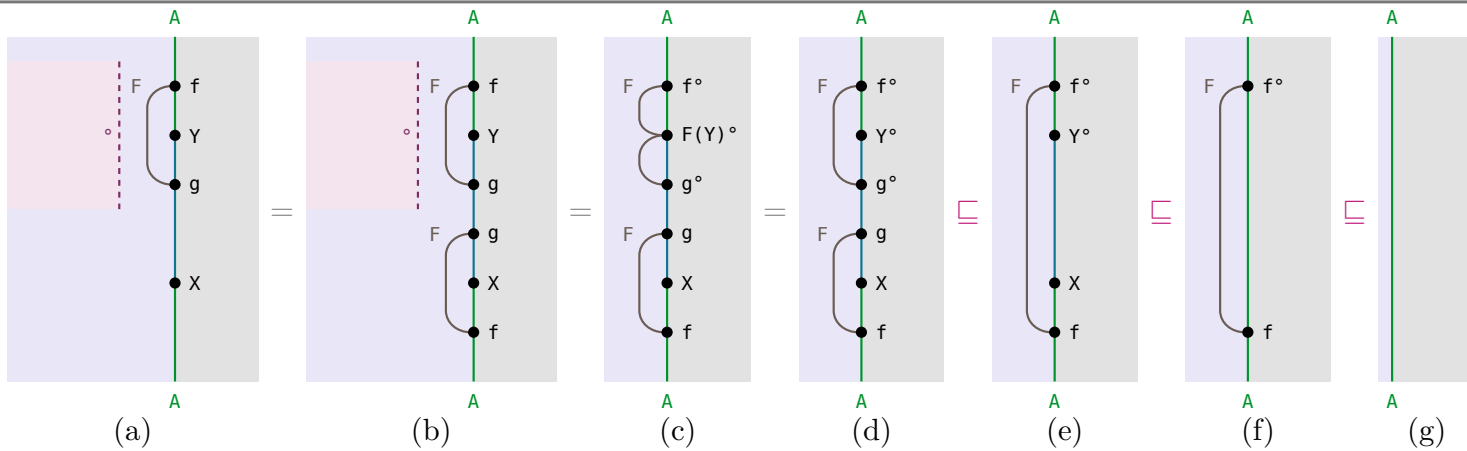
$$\begin{aligned}
 & \mathbf{inductive}(\mathbf{S} \mathbf{member}) \wedge \mathbf{Entire}(\mathbf{S}) \wedge \mathbf{Entire}(\mathbf{R}) \wedge \mathbf{S} \mathbf{F}(\mathbf{X}) \mathbf{R} \sqsubseteq \mathbf{X} \Rightarrow \mathbf{Entire}(\mathbf{X}) & (6.5d) \\
 & \text{when } \mathbf{S} \mathbf{member} \text{ is inductive and } \mathbf{S}, \mathbf{R} \text{ are entire, a pre-fixed point of } \mathbf{X} \mapsto \mathbf{S}\mathbf{F}(\mathbf{X})\mathbf{R} \text{ is entire}
 \end{aligned}$$

$$\mathbf{inductive}(\mathbf{g} \mathbf{member}) \wedge \mathbf{Map}(\mathbf{g}) \wedge \mathbf{Map}(\mathbf{f}) \wedge \mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \Rightarrow \mathbf{Map}(\mathbf{X}) \quad (6.5e)$$

when $\mathbf{g} \mathbf{member}$ is inductive and $\mathbf{f}, \mathbf{g}$ are maps, the solution of $\mathbf{X} = \mathbf{g}\mathbf{F}(\mathbf{X})\mathbf{f}$ is a map

$$\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f} \wedge \mathbf{Simple}(\mathbf{g}) \wedge \mathbf{Simple}(\mathbf{f}) \wedge \mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1} \Rightarrow (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X} \sqsubseteq \mathbf{1}$$

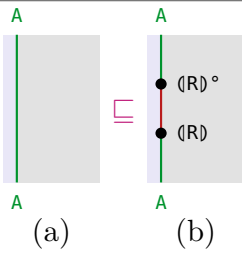
$(\mathbf{g}\mathbf{F}(\mathbf{Y})\mathbf{f})^\circ$ followed by $\mathbf{X}$ stays below $\mathbf{1}$ when $\mathbf{g}, \mathbf{f}$ are simple and $\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$



(a)	$(\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{X}$	$\mathbf{X} = \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$
(b)	$= (\mathbf{g} \mathbf{F}(\mathbf{Y}) \mathbf{f})^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$(\mathbf{R} \mathbf{S})^\circ = \mathbf{S}^\circ \mathbf{R}^\circ$
(c)	$= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y})^\circ \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$\mathbf{PreservesRecip}(\mathbf{F})$
(d)	$= \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ) \mathbf{g}^\circ \mathbf{g} \mathbf{F}(\mathbf{X}) \mathbf{f}$	$\mathbf{Simple}(\mathbf{g})$
(e)	$\sqsubseteq \mathbf{f}^\circ \mathbf{F}(\mathbf{Y}^\circ \mathbf{X}) \mathbf{f}$	$\mathbf{Y}^\circ \mathbf{X} \sqsubseteq \mathbf{1}$
(f)	$\sqsubseteq \mathbf{f}^\circ \mathbf{f}$	$\mathbf{Simple}(\mathbf{f})$
(g)	$\sqsubseteq \mathbf{1}$	

$$\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R) \quad (6.5f)$$

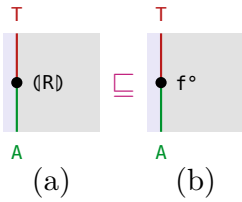
when $R^\circ \text{member}$ is inductive and R is surjective, the fold (R) is surjective



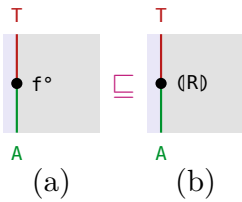
(a)	1	(6.5d) $\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge S \sqsubseteq F(X) R \sqsubseteq X \Rightarrow \text{Entire}(X) \text{ at } S=R^\circ$
(b)	$\sqsubseteq (R)^\circ (R)$	

$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = (R) \quad (6.5g)$$

a map f with $Rf \sqsubseteq F(f) \alpha$, for a surjective R , has the fold (R) as its converse



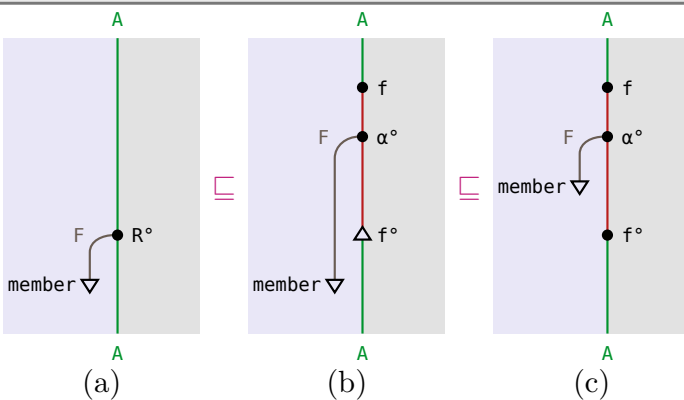
(a)	(R)	shunting f , fusion, $Rf \sqsubseteq F(f) \alpha$
(b)	$\sqsubseteq f^\circ$	



(a)	f°	(6.5f) $\text{inductive}(R^\circ \text{ member}) \wedge 1 \sqsubseteq R^\circ R \Rightarrow 1 \sqsubseteq (R)^\circ (R) \text{ and } (R) f \sqsubseteq 1$
(b)	$\sqsubseteq (R)$	

$$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \Rightarrow R^\circ \text{ member} \sqsubseteq f \alpha^\circ \text{ member } f^\circ \quad (6.5h)$$

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



(a)	$R^\circ \text{ member}$	$\text{Map}(f) \wedge R f \sqsubseteq F(f) \alpha \Rightarrow R^\circ \sqsubseteq f \alpha^\circ F(f^\circ)$
(b)	$\sqsubseteq f \alpha^\circ F(f^\circ) \text{ member}$	$F(R) \text{ member} \sqsubseteq \text{member } R$
(c)	$\sqsubseteq f \alpha^\circ \text{ member } f^\circ$	

§6.6. Sorting by selection

$$\text{ordered} = ([\text{nil}, \text{ok cons}])$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

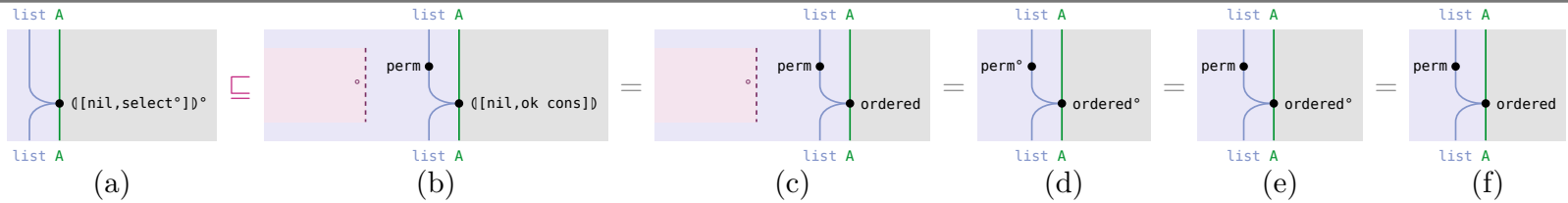


(a)	ordered	fold uniqueness
(b)	= ([nil, ok cons])	

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq \text{perm ordered}$$

(6.6b)

every output of unfolding the input by **select** is a sorted permutation of it

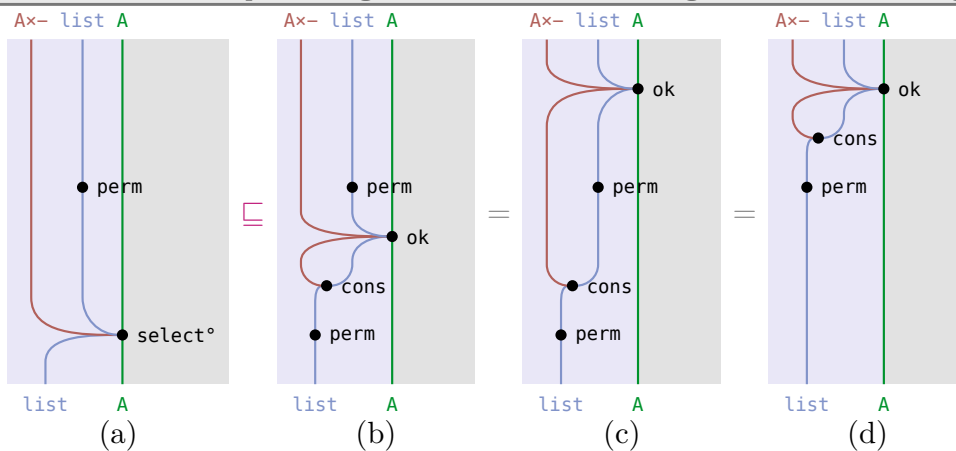


(a)	$[\text{nil}, \text{select}^\circ]^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$ $\Rightarrow ([\text{nil}, \text{select}^\circ])^\circ \sqsubseteq ([\text{nil}, \text{ok cons}])^\circ \text{ perm}^\circ$
(b)	$\sqsubseteq ([\text{nil}, \text{ok cons}])^\circ \text{ perm}^\circ$	(6.6a) $\text{ordered} = ([\text{nil}, \text{ok cons}])^\circ$
(c)	$= (\text{ordered perm})^\circ$	$(R S)^\circ = S^\circ R^\circ$
(d)	$= \text{perm}^\circ \text{ ordered}^\circ$	$\text{perm}^\circ = \text{perm}$
(e)	$= \text{perm ordered}^\circ$	$\text{ordered}^\circ = \text{ordered}$
(f)	$= \text{perm ordered}$	

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{ select}^\circ \sqsubseteq \text{ok cons perm}$$

(6.6c)

permuting the tail and then undoing **select** lands among the **ok** conses of a permutation

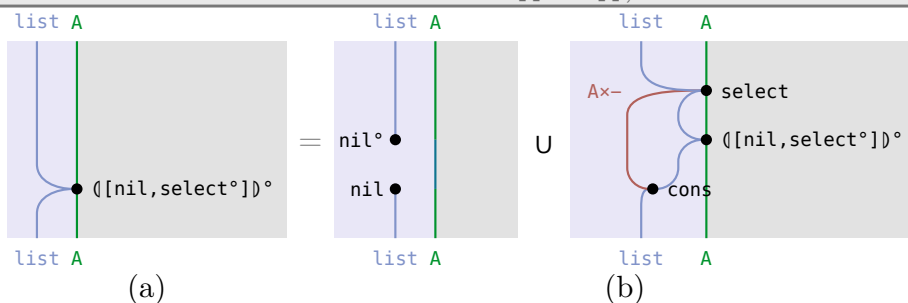


(a)	$(1 \times \text{perm}) \text{ select}^\circ$	$\text{select}^\circ \sqsubseteq \text{ok cons perm}$
(b)	$\sqsubseteq (1 \times \text{perm}) \text{ ok cons perm}$	$\text{ok } (1 \times \text{perm}) = (1 \times \text{perm}) \text{ ok}$
(c)	$= \text{ok } (1 \times \text{perm}) \text{ cons perm}$	$\text{cons perm} = (1 \times \text{perm}) \text{ cons perm}$
(d)	$= \text{ok cons perm}$	

$$([\text{nil}, \text{select}^\circ])^\circ = (\text{nil}^\circ \text{ nil}) \cup (\text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons})$$

(6.6d)

selection sort returns **[]** on **[]**, and otherwise selects **(a, y)**, sorts **y** and conses **a** back on

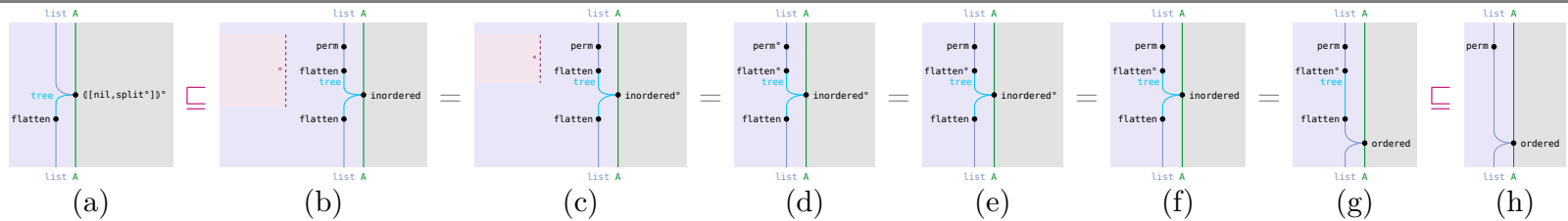


(a)	$[\text{nil}, \text{select}^\circ]^\circ$	unfolding the converse of a fold
(b)	$= (\text{nil}^\circ \text{ nil}) \cup (\text{select } (1 \times ([\text{nil}, \text{select}^\circ])^\circ) \text{ cons})$	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow ([\text{nil}, \text{split}^\circ])^\circ \quad (6.6e)$$

flatten $\sqsubseteq$ perm ordered

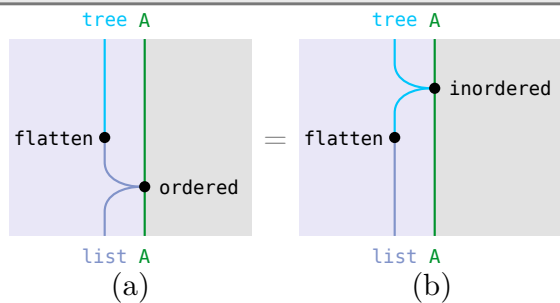
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



(a)	$[\text{nil}, \text{split}^\circ]^\circ \text{ flatten}$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$ $\Rightarrow [\text{nil}, \text{split}^\circ]^\circ \text{ flatten} \sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$
(b)	$\sqsubseteq (\text{inordered flatten perm})^\circ \text{ flatten}$	$(R \ S)^\circ = S^\circ R^\circ$
(c)	$= (\text{flatten perm})^\circ \text{ inordered}^\circ \text{ flatten}$	$(R \ S)^\circ = S^\circ R^\circ$
(d)	$= \text{perm}^\circ \text{ flatten}^\circ \text{ inordered}^\circ \text{ flatten}$	$\text{perm}^\circ = \text{perm}$
(e)	$= \text{perm flatten}^\circ \text{ inordered}^\circ \text{ flatten}$	$\text{inordered}^\circ = \text{inordered}$
(f)	$= \text{perm flatten}^\circ \text{ inordered flatten}$	$(6.6f) (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c))$ $\Rightarrow \text{flatten ordered} = \text{inordered flatten}$
(g)	$= \text{perm flatten}^\circ \text{ flatten ordered}$	$\text{flatten}^\circ \text{ flatten} \sqsubseteq 1$
(h)	$\sqsubseteq \text{perm ordered}$	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

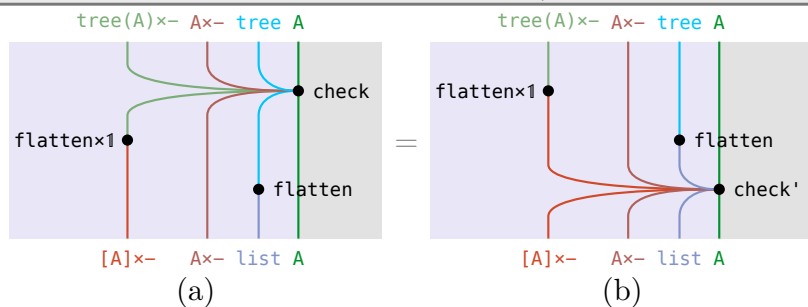
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



(a)	flatten ordered	R transitive: a flattened tree is ordered iff every node passes check
(b)	$= \text{inordered flatten}$	

$$\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{ check}' \quad (6.6g)$$

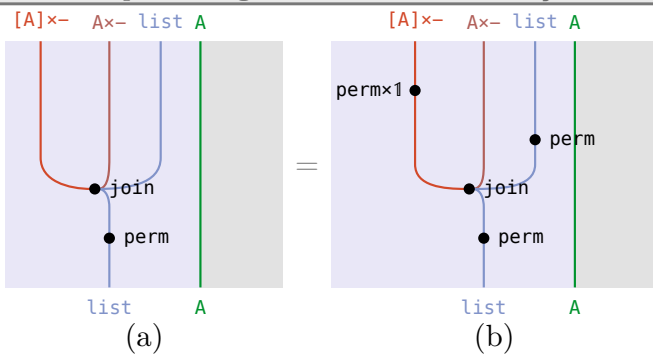
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



(a)	$\text{check} (\text{flatten} \times (1 \times \text{flatten}))$	$b \text{ intree } x \iff b \text{ inlist } \text{flatten}(x)$
(b)	$= (\text{flatten} \times (1 \times \text{flatten})) \text{ check}'$	

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{ join perm} \quad (6.6h)$$

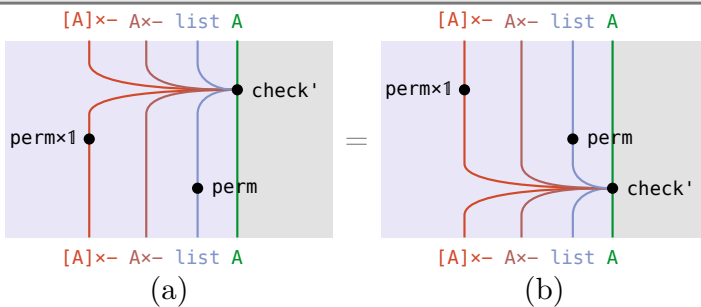
permuting the list $x \mathrel{++} [a] \mathrel{++} y$ is the same as first permuting x and y and then permuting the joined list



(a)	join perm	perm is transitive and closed under ++
(b)	= (perm × (1 × perm)) join perm	

$$\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{check}' \quad (6.6i)$$

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

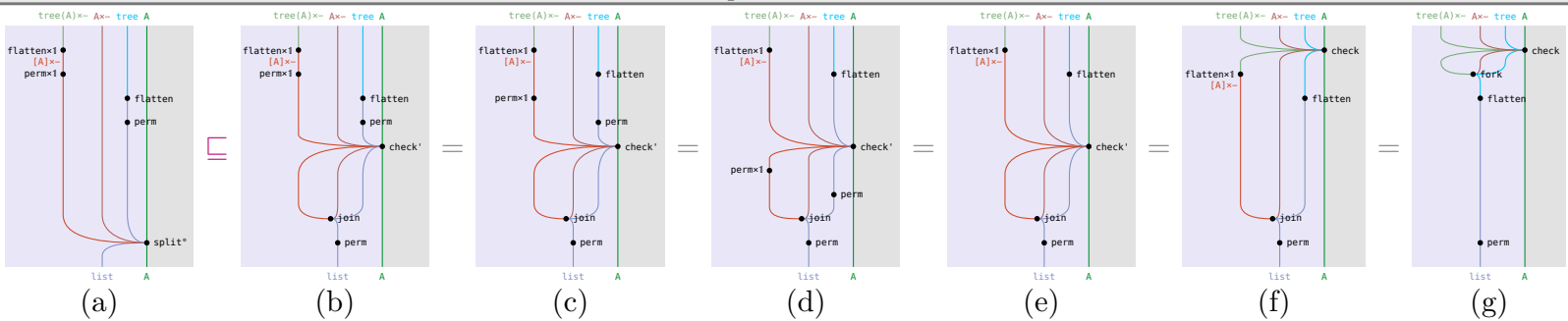


(a)	check' (perm × (1 × perm))	$b \text{ inlist } x \iff b \text{ inlist perm}(x)$
(b)	= (perm × (1 × perm)) check'	

$$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \implies ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ \sqsubseteq \text{check fork} \quad (6.6j)$$

flatten perm

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



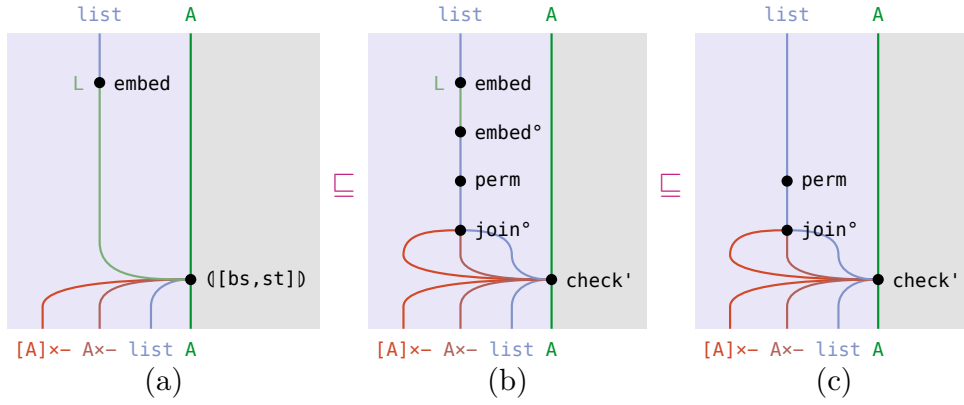
(a)	$((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ$	$\text{split}^\circ \sqsubseteq \text{check}' \text{ join perm}$
(b)	$\sqsubseteq ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{check}' \text{ join perm}$	
(c)	$= (\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{check}' \text{ join perm}$	(6.6i) $\text{check}' (\text{perm} \times (1 \times \text{perm})) = (\text{perm} \times (1 \times \text{perm})) \text{check}'$
(d)	$= (\text{flatten} \times (1 \times \text{flatten})) \text{check}' (\text{perm} \times (1 \times \text{perm})) \text{join perm}$	(6.6h) $\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{join perm}$
(e)	$= (\text{flatten} \times (1 \times \text{flatten})) \text{check}' \text{ join perm}$	(6.6g) $\text{check} (\text{flatten} \times (1 \times \text{flatten})) = (\text{flatten} \times (1 \times \text{flatten})) \text{check}'$
(f)	$= \text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{join perm}$	$\text{fork flatten} = (\text{flatten} \times (1 \times \text{flatten})) \text{join}$
(g)	$= \text{check fork flatten perm}$	

$bs \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed}$ (6.6k)

$([bs, st]) \sqsubseteq \text{perm join}^\circ \text{check}'$

turning a list into a non-empty list and folding it with any $[bs, st]$ meeting the two conditions below gives only

checked triples (x, a, y) whose $\text{join } x \# [a] \# y$ is a permutation of the list

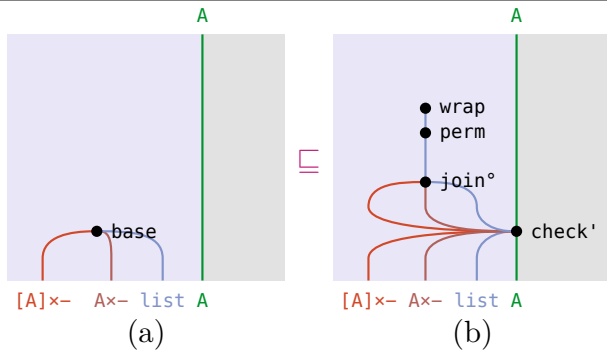


(a)	$\text{embed } ([bs, st])$	$bs \sqsubseteq \text{wrap perm join}^\circ \text{check}' \wedge (1 \times (\text{perm join}^\circ \text{check}')) \text{ st} \sqsubseteq \text{cons perm join}^\circ \text{check}' \Rightarrow \text{embed } ([bs, st]) \sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$
(b)	$\sqsubseteq \text{embed embed}^\circ \text{perm join}^\circ \text{check}'$	$\text{embed embed}^\circ \text{perm join}^\circ \text{check}' \sqsubseteq \text{perm join}^\circ \text{check}'$
(c)	$\sqsubseteq \text{perm join}^\circ \text{check}'$	

$\text{base} \sqsubseteq \text{wrap perm join}^\circ \text{check}'$

(6.6l)

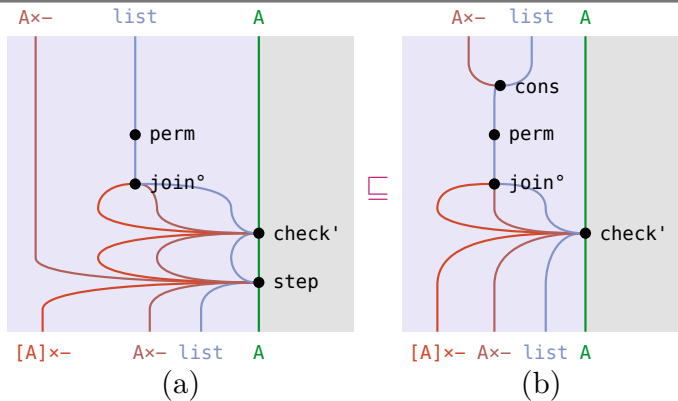
$\text{base}(a)$ is a **checked triple** whose join is a permutation of the one-element list $[a]$



(a)	base	$[] \# [a] \# [] = [a]$, and check' holds on empty lists
(b)	$\sqsubseteq \text{wrap perm join}^\circ \text{check}'$	

$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ \text{check}')) \text{ step} \sqsubseteq \text{cons perm join}^\circ \text{check}'$ (6.6m)

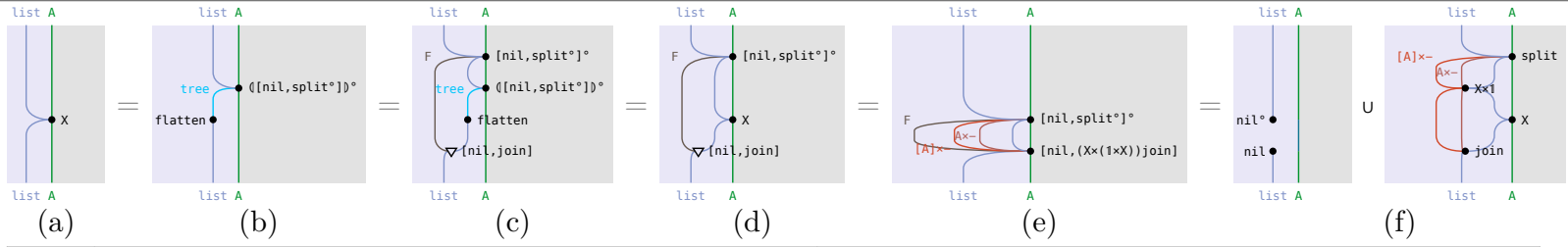
if (x, b, y) is a **checked triple** for a permutation of l , then $\text{step}(a, (x, b, y))$ is a **checked triple** for a permutation of $[a] \# l$



(a)	$(1 \times (\text{perm join}^\circ \text{check}')) \text{ step}$	aRb puts a in front of x ; otherwise bRa and a goes in front of y
(b)	$\sqsubseteq \text{cons perm join}^\circ \text{check}'$	

$$X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} \Rightarrow X = (\text{nil}^\circ \text{ nil}) \cup (\text{split} (X \times (1 \times X)) \text{ join}) \quad (6.6n)$$

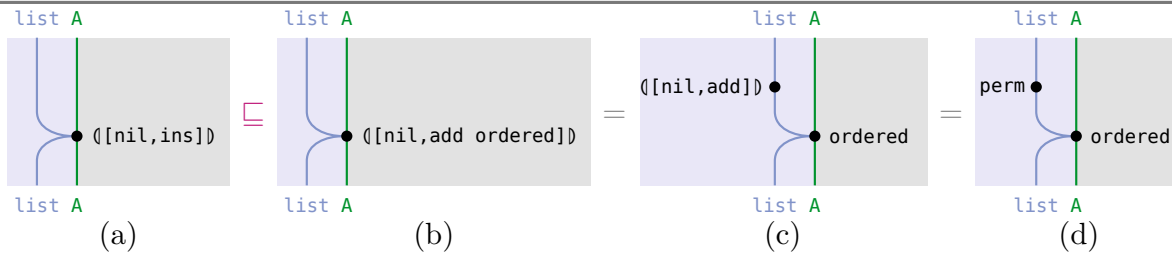
quicksort returns [] on [], and otherwise splits into (x,a,y), sorts x and y and joins them around a



(a)	X	$X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$
(b)	$= ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$	$S^\circ F((S)^\circ (R)) R = (S)^\circ (R)$
(c)	$= [\text{nil}, \text{split}^\circ]^\circ F([\text{nil}, \text{split}^\circ])^\circ \text{ flatten} [\text{nil}, \text{join}]$	$X = ([\text{nil}, \text{split}^\circ])^\circ \text{ flatten}$
(d)	$= [\text{nil}, \text{split}^\circ]^\circ F(X) [\text{nil}, \text{join}]$	$F(S) [T, U] = [T, (S \times (1 \times S)) U]$
(e)	$= [\text{nil}, \text{split}^\circ]^\circ [\text{nil}, (X \times (1 \times X)) \text{ join}]$	$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S)$
(f)	$= (\text{nil}^\circ \text{ nil}) \cup (\text{split} (X \times (1 \times X)) \text{ join})$	

$$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow ([\text{nil}, \text{ins}])^\circ \sqsubseteq \text{perm ordered} \quad (6.6o)$$

for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations

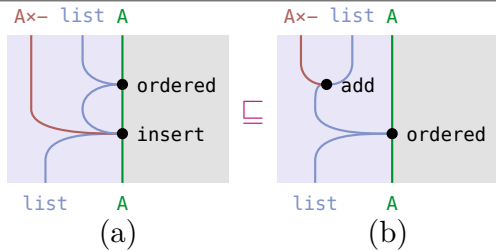


(a)	$([\text{nil}, \text{ins}])^\circ$	$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered}$
(b)	$\sqsubseteq ([\text{nil}, \text{add ordered}])^\circ$	$\Rightarrow ([\text{nil}, \text{ins}])^\circ \sqsubseteq ([\text{nil}, \text{add ordered}])^\circ$
(c)	$= ([\text{nil}, \text{add}])^\circ \text{ ordered}$	$([\text{nil}, \text{add}])^\circ \text{ ordered} = ([\text{nil}, \text{add ordered}])^\circ$
(d)	$= \text{perm ordered}$	$\text{perm} = ([\text{nil}, \text{add}])^\circ$

$$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \wedge (\forall a. \forall b. \forall c. \quad (6.6p)$$

$$R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \Rightarrow (1 \times \text{ordered}) \text{ insert} \sqsubseteq \text{add ordered}$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



(a)	$(1 \times \text{ordered}) \text{ insert}$	insert splices a in, and R transitive and total
(b)	$\sqsubseteq \text{add ordered}$	

§6.7. Closure

$$R^* = (\mu X : 1 \cup (X R)) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R***

$$1 \sqsubseteq X \wedge X X \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R***

$$1 \sqsubseteq R^* \quad (6.7c)$$

R* is reflexive

(a)	1	$R \sqsubseteq R \cup S$
(b)	$\sqsubseteq 1 \cup (R R^*)$	$1 \cup (R R^*) = R^*$
(c)	$= R^*$	

$$R \subseteq R^*$$

(6.7d)

 R^* contains R

(a)	R	
(b)	$= R$	(6.7c) $1 \subseteq R^*$
(c)	$\sqsubseteq R R^*$	$R R^* \subseteq R^*$
(d)	$\sqsubseteq R^*$	

$$(1 \cup (R (R^* / R^*))) R^* \subseteq R^*$$

(6.7e)

 R^*/R^* is a prefixed point of $X \mapsto 1 \cup R X$, stated through division

$$R^* R^* \subseteq R^*$$

 R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto 1 \cup R X$

(a)	$(1 \cup (R (R^* / R^*))) R^*$	$(S \cup T) R = (S R) \cup (T R)$
(b)	$= (R^*) \cup (R (R^* / R^*) R^*)$	
(c)	$= R^* \cup (R (R^* / R^*) R^*)$	$(R / R) R \subseteq R$
(d)	$\sqsubseteq R^* \cup (R R^*)$	$R R^* \subseteq R^*$
(e)	$\sqsubseteq R^*$	

$$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \Rightarrow 1 \cup (R X) \subseteq X$$

(6.7f)

a reflexive transitive X containing R is a prefixed point of $Y \mapsto 1 \cup R Y$

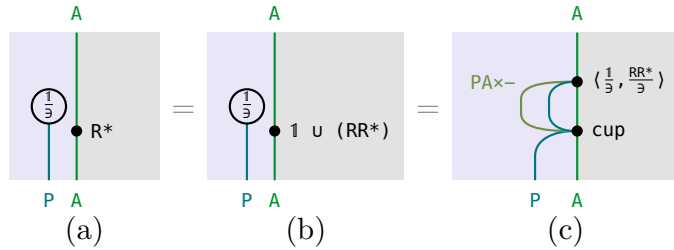
$$1 \subseteq X \wedge X X \subseteq X \wedge R \subseteq X \Rightarrow R^* \subseteq X$$

a reflexive transitive X containing R is a prefixed point of $X \mapsto 1 \cup R X$, so it contains R^*

(a)	$1 \cup (R X)$	$1 \subseteq X$
(b)	$\sqsubseteq X \cup (R X)$	$R \subseteq X$
(c)	$\sqsubseteq X \cup (X X)$	
(d)	$\sqsubseteq X$	

$$\frac{R^*}{\exists} = \left(\frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$$

(6.7g)

the set of R^* -successors of a is a itself joined with the R^* -successors of its R -successors


(a)	$\frac{R^*}{\exists}$	$1 \cup (R R^*) = R^*$
(b)	$= \frac{1 \cup (R R^*)}{\exists}$	$\frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{ cup}$
(c)	$= \left(\frac{1}{\exists}, \frac{R R^*}{\exists} \right) \text{ cup}$	

$$R - \emptyset = R$$

(6.7h)

taking nothing away leaves R

$$R \cup (S - R) = R \cup S$$

beside R , only the part of S outside R adds anything

$$R - (S \cup T) = R - S - T$$

taking away S and then T takes away $S \cup T$

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking T away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \Rightarrow (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$$

(6.7i)

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

$$S R^* = (\mu X : S \cup (X R))$$

(6.7j)

 S followed by any number of R steps is the least X containing S and closed under a further R step

$$R^* S = (\mu X : S \cup (R X))$$

any number of R steps followed by S is the least X containing S and closed under an R step in front

$$\theta(\emptyset, S) = S R^*$$

(6.7k)

started with nothing found and S to explore, θ computes SR^*

(a)	$\theta(\emptyset, S)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
(b)	$\triangleq \emptyset \cup (\mu X : S \cup ((X R) - \emptyset))$	(6.7h) $R - \emptyset = R$
(c)	$= \emptyset \cup (\mu X : S \cup (X R))$	$\emptyset \cup R = R$
(d)	$= (\mu X : S \cup (X R))$	(6.7j) $S R^* = (\mu X : S \cup (X R))$
(e)	$= S R^*$	

$$\theta(P, \emptyset) = P$$

(6.7l)

with nothing left to explore, θ returns what it has found

(a)	$\theta(P, \emptyset)$	$\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((X R) - P))$
(b)	$\triangleq P \cup (\mu X : \emptyset \cup ((X R) - P))$	$(\mu X : \emptyset \cup ((X R) - P)) = \emptyset$
(c)	$= P \cup \emptyset$	$R \cup \emptyset = R$
(d)	$= P$	